Zip Linked List

AlgoExpert · Very Hard · Linked Lists

The problem

- Reorder in place to first, last, second, second-to-last, … Split at middle, reverse second half, interleave.

Intuition

- Split at the middle, reverse the back half, and interleave.

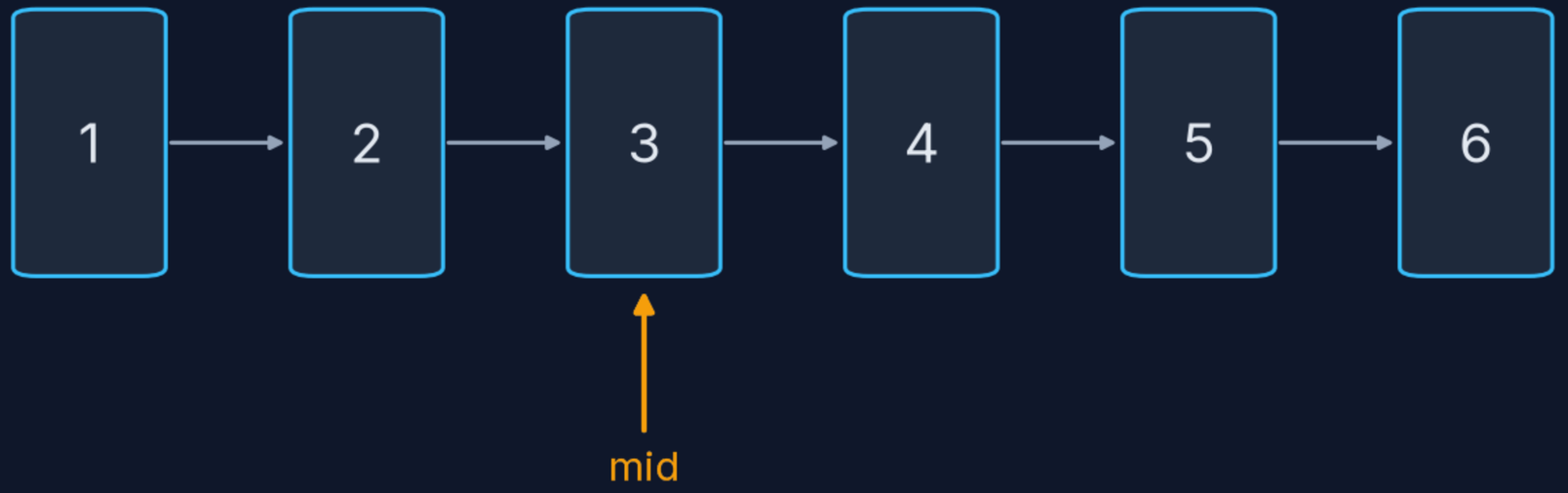
Brute force vs optimal

- Array of nodes, two pointers → O(n) space
- Middle + reverse + merge → O(1)

Approach

- 1. Find the middle
- 2. Reverse the second half
- 3. Alternate nodes from both halves

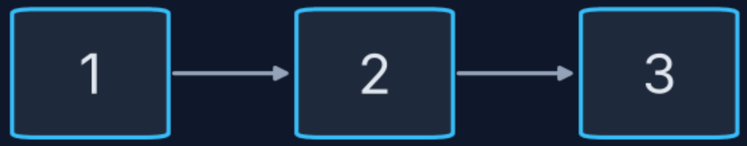
Worked example



Find the middle

Worked example

first



second reversed



Split and reverse the second half

Worked example



Interleave

Java solution

```java
public class ZipLinkedList {
    public static LinkedListNode zipLinkedList(LinkedListNode head) {
        if (head == null || head.next == null) return head;
        LinkedListNode slow = head, fast = head;
        while (fast.next != null && fast.next.next != null) { slow = slow.next; fast = fast.next.next; }
        LinkedListNode second = slow.next, prev = null;
        slow.next = null;
        while (second != null) { LinkedListNode n = second.next; second.next = prev; prev = second; second = n; }
        LinkedListNode a = head, b = prev;
        while (b != null) {
            LinkedListNode an = a.next, bn = b.next;
            a.next = b; b.next = an;
            a = an; b = bn;
        }
        return head;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Cut the first half's tail
- Odd lengths
- Short lists

Interview tips & follow-ups

- Same as Reorder List

Recap

- Split at the middle, reverse the back half, and interleave.
- Complexity: O(n)